Question		Answer	Marks	AO	Guidance
9	(a)	$\begin{pmatrix}3\\3\\-3\end{pmatrix}$ **or** $3\mathbf{i}+3\mathbf{j}-3\mathbf{k}$	**B1** **[1]**	**1.1**	do not allow (3, 3, −3); mark the final answer
9	(b)	$[\overrightarrow{OC}=]\begin{pmatrix}-1\\-2\\0\end{pmatrix}+\frac{1}{3}\begin{pmatrix}3\\3\\-3\end{pmatrix}$ **oe** $\begin{pmatrix}0\\-1\\-1\end{pmatrix}$ **or** $-\mathbf{j}-\mathbf{k}$	**M1** **A1**	**3.1a** **1.1**	**FT** their $\begin{pmatrix}3\\3\\-3\end{pmatrix}$ allow **B2** for correct answer unsupported; mark the final answer if **M0** allow **SC1** for $\left[\begin{pmatrix}-1\\-2\\0\end{pmatrix}+\frac{2}{3}\begin{pmatrix}3\\3\\-3\end{pmatrix}=\right]\begin{pmatrix}1\\0\\-2\end{pmatrix}$ do not allow (0, −1, −1); only penalise coordinate form once
		Alternative method 1 $[\overrightarrow{AC}=]\begin{pmatrix}a--1\\b--2\\c-0\end{pmatrix}$ $[\overrightarrow{CB}=]\begin{pmatrix}2-a\\1-b\\-3-c\end{pmatrix}$ $2\overrightarrow{AC}=\overrightarrow{CB}$ equating coefficients: $2a+2=2-a$ $2b+4=1-b$ $2c=-3-c$ $\begin{pmatrix}0\\-1\\-1\end{pmatrix}$ **or** $-\mathbf{j}-\mathbf{k}$	 **M1** **A1**		may work with **i, j, k** notation allow one sign error only do not allow (0, −1, −1); only penalise coordinate form once

Question		Answer	Marks	AO	Guidance
		Alternative method 2 $2\overrightarrow{OC} - 2\begin{pmatrix}-1\\-2\\0\end{pmatrix} = \begin{pmatrix}2\\1\\-3\end{pmatrix} - \overrightarrow{OC}$ $\begin{pmatrix}0\\-1\\-1\end{pmatrix}$ **or** $-\mathbf{j} - \mathbf{k}$	**M1** **A1**		may see $3\overrightarrow{OC} = \begin{pmatrix}0\\-3\\-3\end{pmatrix}$; allow one sign error only; may work with **i, j, k** notation do not allow (0, −1, −1); only penalise coordinate form once
		Alternative method 3 $[\overrightarrow{OC} =]\frac{2}{3}\begin{pmatrix}-1\\-2\\0\end{pmatrix} + \frac{1}{3}\begin{pmatrix}2\\1\\-3\end{pmatrix}$ $\begin{pmatrix}0\\-1\\-1\end{pmatrix}$ **or** $-\mathbf{j} - \mathbf{k}$	**M1** **A1**		ratio theorem; may work with **i, j, k** notation do not allow (0, −1, −1); only penalise coordinate form once
			[2]		